



UNIVERSITY OF
WATERLOO

MTE 203 – Advanced Calculus

Midterm Exam Fall 2020 – Problem 4

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If $u(x, y) = x^2 f\left(\frac{y}{x}\right)$, where f is a differentiable function, use the chain rule to show that $u(x, y)$ satisfies,

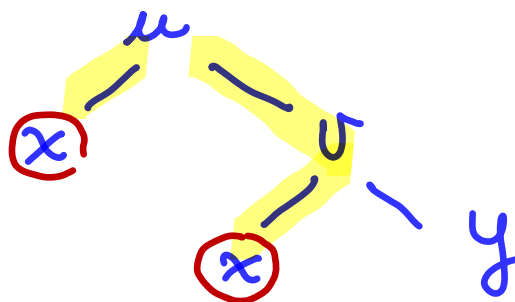
$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$$

$\textcircled{\text{I}} \quad \textcircled{\text{II}}$

Solution:

$$\text{LET } v = \frac{y}{x} \quad \rightarrow \quad u = x^2 f(v)$$

TREE DIAGRAM :



TWO BRANCHES

Index: Chain Rule



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Midterm Exam Fall 2020 – Problem 4

$$\textcircled{\text{I}} \quad \frac{\partial u}{\partial x} = ?$$

$$\frac{\partial u}{\partial x} = \underbrace{\left. \frac{\partial u}{\partial x} \right|_v}_{\textcircled{\text{A}}} + \underbrace{\frac{\partial u}{\partial v} \cdot \frac{\partial v}{\partial x}}_{\textcircled{\text{B}}}$$

$$\textcircled{\text{A}} \quad \left. \frac{\partial u}{\partial x} \right|_v = \frac{\partial}{\partial x} (x^2 f(v)) \Big|_v = 2x f(v)$$

$$\textcircled{\text{B}} \quad \frac{\partial u}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial}{\partial v} (x^2 f(v)) \cdot \frac{\partial v}{\partial x}$$



Midterm Exam Fall 2020 – Problem 4

$$= x^2 \frac{\partial f(u)}{\partial u} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right)$$

$$= \cancel{x^2} \frac{\partial f(u)}{\partial u} \cdot \left(-\frac{y}{\cancel{x^2}} \right)$$

$$= - \frac{\partial f(u)}{\partial u} y$$

$$\Rightarrow \frac{\partial u}{\partial x} = \underbrace{2x f(u)}_{\textcircled{A}} - y \frac{\partial f(u)}{\partial u} \dots \textcircled{1}$$



Midterm Exam Fall 2020 – Problem 4

$$\textcircled{\text{II}} \quad \frac{\partial u}{\partial y} = ?$$

PLEASE TRY TO SOLVE THIS PART ON
YOUR OWN FIRST !

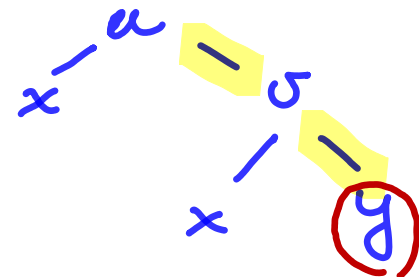
SOLUTION CONTINUES
ON NEXT PAGE 



Midterm Exam Fall 2020 – Problem 4

(II) $\frac{\partial u}{\partial y} = ?$

SAME
TREE $\rightarrow$



ONE BRANCH

$$\frac{\partial u}{\partial y} = \underbrace{\frac{\partial u}{\partial v}}_C \underbrace{\frac{\partial v}{\partial y}}_D$$

(C) $\frac{\partial u}{\partial v} = \frac{\partial}{\partial v} (x^2 f(v)) = x^2 \frac{\partial f(v)}{\partial v}$

(D) $\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \left(\frac{y}{x} \right) = \frac{1}{x}$



Midterm Exam Fall 2020 – Problem 4

$$\rightarrow \frac{\partial u}{\partial y} = x^2 \frac{\partial f(u)}{\partial v} \left(\frac{1}{x} \right)$$

$$\frac{\partial u}{\partial y} = x \frac{\partial f(u)}{\partial v} \quad \dots (2)$$

(1) AND (2) INTO MAIN EQUATION,

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= x \left[2x f(u) - y \frac{\partial f(u)}{\partial v} \right] + y \left[x \frac{\partial f(u)}{\partial v} \right] \\ &= \underbrace{2x^2 f(u)}_u - \cancel{xy \frac{\partial f(u)}{\partial v}} + \cancel{xy \frac{\partial f(u)}{\partial v}} \end{aligned}$$

$$\rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$$

$\rightarrow$ QED



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